

THE IDEA

- Replace MLD's text conditioning with the vehicle ego trajectory.
- Feed the denoiser ALL 196 per-timestep ego tokens — not one pooled summary — so every denoising step has temporal resolution over the trajectory.
- Ego encoder is contrastively pretrained, then FROZEN; only the denoiser trains.

MAIN RESULTS · UNIFIED EVALUATION, 3 REPLICATIONS

| System                              | FID ↓ | R@1   | MM   | Div  |
|-------------------------------------|-------|-------|------|------|
| EgoPed-IA (+ interaction sampling)  | 2.880 | 0.323 | 3.37 | 5.67 |
| EgoPed (full-sequence conditioning) | 3.392 | 0.548 | 3.96 | 5.79 |
| Unfrozen encoder                    | 5.079 | 0.352 | 2.62 | 6.06 |
| Pooled-token baseline               | 5.620 | 0.665 | 3.71 | 5.82 |
| Larger latent (8 vs 4)              | 6.677 | 0.663 | 3.37 | 5.89 |
| Ground truth                        | —     | —     | —    | 5.50 |

ARCHITECTURE VS. DATA CURATION · 2×2

| FID ↓         | – interaction sampling | + interaction sampling |
|---------------|------------------------|------------------------|
| Pooled        | 6.214                  | 5.620                  |
| Full-sequence | 3.392                  | 2.880                  |

Conditioning granularity dominates (≈2.7 FID in both columns, ~5× the data effect); the two effects are additive → the headline is conservative, not inflated.

**RESULT:** information ladder (FID / ADE / behaviour-separation) — 0 bits: 3.24 / 2.99 / 0.04 → 5-bit vehicle text: 5.91 / 2.52 / 0.21 → oracle text (told the behaviour): 2.82 / 1.90 / 0.93 → trajectory (H4): 3.39 / 2.32 / 0.33. The trajectory recovers 61% of the know-nothing→know-the-outcome ADE gap, vehicle text 43%. FID ranks marginals, not conditioning — an ego-BLIND model trained on IA's interaction-sampling recipe scores the best FID in the study (1.43) at chance-level conditioning (ADE 3.00, separation 0.02), so IA's FID lead is data curation, not conditioning; H4 is the best-conditioned model. A second IA seed replicates both halves: FID 2.671 vs 2.880, and the same per-condition shortfall (ADE 2.61, separation 0.14). **Open:** pseudo GT is a canonical body that SLIDES (skate ratio 0.07; one skeleton for all 14,173) — most trajectories are sound, body pose is not — no physics constraints — no perceptual study yet.

WHAT ACTUALLY MATTERS

| Question                          | Finding           | Verdict                  |
|-----------------------------------|-------------------|--------------------------|
| Unfreeze the ego encoder?         | MM 3.96→2.62      | freezing critical        |
| Skip encoder pretraining?         | R@1 = chance      | ≈ unconditional          |
| Cross- instead of self-attention? | 3.49 vs 3.34      | no difference (p=.33)    |
| Bigger VAE latent (8 vs 4)?       | 6.68 vs 5.62      | 19% worse                |
| Different training seed?          | H4 3.39/4.72/4.80 | H4 varies, IA replicates |

EXTERNAL BASELINES · CAPABILITY LADDER

| Approach                         | FID ↓ | Reading                             |
|----------------------------------|-------|-------------------------------------|
| Uncond. + interaction sampling   | 1.43  | BEST FID — and fully ego-blind      |
| EgoPed-IA (ours)                 | 2.88  | in-domain, ego-conditioned          |
| Unconditional, properly trained  | 3.24  | ≈ H4 — FID is blind to conditioning |
| Unconditional prior (ego zeroed) | 5.18  | a weak null branch, not a prior     |
| Pretrained text-to-motion MLD    | 14.07 | T2M does not transfer               |
| Raw-space diffusion (MDM-style)  | 26.43 | latent backbone justified           |
| Ego→motion regressor             | 35.47 | L2-optimal, unrealistic             |
| Trajectory + static body         | 47.7  | needs articulated motion            |

VALIDATED ON FOUR INDEPENDENT AXES

- FID — 40% over pooled (non-overlapping CIs); but a trained prior reaches 3.24 and, on IA's own data recipe, 1.43 — better than every conditional model. FID alone does not credit conditioning.
- Held-out split — scene-disjoint, never used for selection: ranking preserved.
- Trajectory ADE/FDE — no learned evaluator: H4 beats pooled on all 4 metrics; a trained prior with the SAME FID is 29% worse (ego-blind) — and EgoPed-IA, best by FID, is worse than H4 per-condition (ADE 2.64 vs 2.32; separation 0.15 vs 0.33; still behind at matched epochs); H4 is the best-conditioned model.